

Project Scoping

Name: Sports Car Price Analysis and Market Trends.

Business Problem: The sports car industry is extremely competitive and price-sensitive, with pricing varying greatly between brands, models, and performance parameters. Consumers and organizations frequently lack clarity about how parameters such as horsepower, engine size, torque, and acceleration affect cost. This complicates making informed purchasing decisions, establishing competitive pricing, and understanding market positioning. The challenge is to determine which performance variables have the most impact on sports car prices and how these connections differ between manufacturers and models.

Business Impact: The analysis will provide valuable insights for a variety of stakeholders. Consumers will benefit from knowing whether increased prices are justified by performance characteristics. Automotive firms and dealerships can utilize the information to improve pricing and product placement. Furthermore, this project can boost decision-making efficiency by identifying key pricing determinants and enabling predictive pricing models. Overall, the effort intends to improve accuracy and transparency in determining the value of sports cars.

Dataset(s):

Primary Dataset: Sports Car Prices Dataset.

Contains variables such as car make, model, year, engine size, horsepower, torque, acceleration (0–60 MPH), and price.

Source: <https://www.kaggle.com/datasets/rkiattisak/sports-car-prices-dataset>

Additional Datasets (for validation):

Car Price Analysis Dataset.

Vehicle Price Prediction Dataset.

Source: <https://www.kaggle.com/datasets/ayeshasiddiqa123/cars-pre/data?utm>

https://baselight.app/u/kaggle/dataset/metawave_vehicle_price_prediction?utm

Strengths:

- Includes key performance metrics directly related to pricing.
- Structured and easy to analyze.
- Suitable for regression and correlation analysis.

Weaknesses:

- Primary dataset is LLM-generated (not real-world data).
- May lack real market variability and external factors (e.g., demand, brand perception).
- Limited size compared to large real-world datasets.

Methods:**Variables and Comparisons:**

Independent Variables: Engine Size, Horsepower, Torque, 0–60 MPH Time, Year.

Dependent Variable: Price.

Analysis Approach:

- Exploratory Data Analysis (EDA) to identify patterns and outliers.
- Correlation analysis to measure relationships between variables.
- Regression analysis predicts prices.
- Data visualization to interpret trends.

Dashboard:

The final dashboard will include:

- Scatter plots showing the relationship between horsepower and price.
- Filters for car make and year.
- Bar charts compare average prices across brands.
- Heatmap showing correlation between variables.
- KPI cards display average price, max price, and performance ranges.

Milestones:

- Data Collection and Cleaning.
- Exploratory Data Analysis (EDA).
- Data Visualization Development.

- Statistical Analysis (Correlation and Regression).
- Dashboard Creation.
- Final Report and Presentation.

Timeline:

Week	Tasks
Week 1	Understand dataset and define the problem. Data cleaning and preprocessing. Perform EDA. Generate initial visualizations.
Week 2	Conduct correlation and regression analysis. Begin building dashboard.
Week 3	Finalize the dashboard. Validate findings using additional datasets. Complete final report.